

**An Internship Report**  
Of  
**C++ Programming Internship**

Submitted  
To

**School of Computer Science and Engineering**  
**IILM University, Rajasthan**

In partial fulfilment of the requirement of degree of  
**B.Tech (Computer Science and Engineering)**



By

Roll Number of Student: 25SCS1003002397

**Name of Student: DINESH KUMAR**

Section: 2CSE24

Batch: 2025-2029

## **CANDIDATE'S DECLARATION**

I, DINESH KUMAR, do hereby declare that the internship report titled "C++ Programming Internship" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:-

Student Name: DINESH KUMAR

Roll No.: 25SCS1003002397

Section: 2CSE24

## **ACKNOWLEDGEMENT**

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank the team at Thiranex for providing me the opportunity and industry mentorship to gain hands-on, practical exposure in C++ Programming through a structured, project-based internship. I am also thankful to my institute, IILM University, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement.

I express my gratitude to my parents for their blessings.

Signature:-

Student Name: DINESH KUMAR

Roll No.: 25SCS1003002397

# TABLE OF CONTENTS

|                                                               |          |
|---------------------------------------------------------------|----------|
| 1. Candidate's Declaration .....                              | i        |
| 2. Acknowledgement .....                                      | ii       |
| 3. Internship Offer Letter .....                              | 1        |
| <b>4. Project Description</b> .....                           | <b>2</b> |
| 4.1 Introduction .....                                        | 2        |
| 4.2 Organization Profile .....                                | 2        |
| 4.3 Problem Statement .....                                   | 3        |
| 4.4 Project Objectives .....                                  | 3        |
| 4.5 Scope of the Project .....                                | 3        |
| 4.6 Technologies and Tools Used .....                         | 3        |
| 4.7 System Architecture .....                                 | 3        |
| 4.8 Methodology .....                                         | 3        |
| 4.9 Expected Outcomes .....                                   | 4        |
| 4.10 Certificates of Completion and Communication Proof ..... | 4        |
| <b>5. Bibliography/References</b> .....                       | <b>5</b> |

### 3. INTERNSHIP OFFER LETTER

The following is a true reproduction of the internship offer letter issued to the candidate by Thiranex.

**Thiranex**  
Skill Development & Future Tech

Date: 31 July 2026

TO,

**Dinesh Kumar**

Intern ID: THX-JUL3126-376

**Subject: Offer Letter for Internship in C++ Programming**

Dear Dinesh Kumar,

Following our selection process, we are pleased to offer you an internship position at Thiranex. We believe your skills and enthusiasm will be a great addition to our team.

|                          |                          |
|--------------------------|--------------------------|
| <b>Internship Role</b>   | Intern - C++ Programming |
| <b>Commencement Date</b> | 31 Jul 2026              |
| <b>Completion Date</b>   | 30 Aug 2026              |
| <b>Work Mode</b>         | Remote / Project-Based   |

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.

This is an electronically generated document. No physical signature is required.  
For verification, contact Thiranex Verification Cell.  
[www.thiranex.in](http://www.thiranex.in)

## **4. PROJECT DESCRIPTION**

### **4.1 Introduction**

This report presents a detailed account of the one-month internship undertaken at Thiranex, in the domain of C++ Programming. The internship was carried out from 31 July 2026 to 30 August 2026 as part of practical, skill-building exposure alongside the requirements of the Bachelor of Technology (Computer Science and Engineering) programme, with a specialization in Graphics and Game Technology, at IILM University. The primary aim of this internship was to build a strong practical foundation in C++ programming — covering object-oriented programming, file handling, and menu-driven console application design — by independently building and submitting a series of complete, working applications under industry mentorship.

The internship followed a project-based structure in which four console-based C++ applications were assigned at weekly intervals, each reviewed for correctness and completeness. On successful, on-track submission of all four projects, the internship period concluded on 30 August 2026 as specified in the offer letter issued by Thiranex.

### **4.2 Organization Profile**

Thiranex is an organization working under the theme "Skill Development & Future Tech," offering structured, industry-mentored internship programs to engineering students. The C++ Programming internship track was delivered in a remote, project-based format, in which interns independently build and submit practical console applications under periodic review, with successful completion determined by the quality and timeliness of project submissions.

### **4.3 Problem Statement**

Although many computer science students learn object-oriented programming, data structures, and file handling as part of academic coursework, very few get structured, hands-on practice combining these concepts into complete, working applications. Concepts such as classes and objects, persistent file-based storage, and menu-driven control flow are often understood only at a theoretical level, with limited practice in designing, testing, and submitting real console applications end-to-end. The problem addressed by this internship was to practically apply core C++ and object-oriented programming concepts by independently designing, coding, and submitting four progressively structured console-based applications.

### **4.4 Project Objectives**

- To apply object-oriented programming concepts — classes, objects, and encapsulation — to real, practical problems.
- To implement file handling for persistent storage of records across program runs.
- To design clear, menu-driven navigation for console-based applications.
- To build a Student Management System capable of adding, updating, deleting, and displaying student records.

- To build a Bank Management Application simulating deposit, withdrawal, and balance-check operations.
- To build a Library Management System supporting book/member management, issue-return processing, and search.
- To build a Mini Game Project (Tic Tac Toe / Snake) demonstrating loops, arrays, and conditional logic.
- To test, debug, and submit each project on schedule under industry mentorship review.

#### 4.5 Scope of the Project

The scope of this internship spans four independent but conceptually progressive C++ console applications, each focused on a different application domain — student record management, banking operations, library management, and game logic — while reinforcing a common set of core programming skills: object-oriented design, file handling, and structured, menu-driven control flow. The internship was limited to a remote, project-based learning environment provided by Thiranex, and did not involve deployment on any live organizational production system.

#### 4.6 Technologies and Tools Used

- **Programming Language:** C++
- **Programming Paradigm:** Object-Oriented Programming (classes, objects, encapsulation)
- **Data Persistence:** File handling for reading and writing records
- **Data Organization:** Arrays and structures for in-memory record management
- **Program Control:** Loops and conditional statements for control and game logic
- **Interface Design:** Menu-driven console navigation
- **Development Practice:** Manual testing and debugging of console applications

#### 4.7 System Architecture

The common architecture followed across all four C++ console applications built during the internship can be summarized in the following layers:

- **Input Layer:** Console-based user input captured via menu-driven prompts.
- **Logic Layer:** Core program logic implemented using object-oriented principles — classes representing students, accounts, books, or game state.
- **Data Layer:** Persistent storage and retrieval of records using file handling.
- **Control Layer:** Loops, conditionals, and validation or win/loss logic driving each application's behaviour.
- **Output Layer:** Formatted console output, including dynamically updated displays such as record listings or the game board.

#### 4.8 Methodology

The internship followed a task-based methodology, with one console application assigned and reviewed each week, as summarized in the table below.

| Task | Project                                 | Description                                                                                                    | Due Date    | Status                         |
|------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------|--------------------------------|
| 1    | Student Management System               | Console-based system to add, update, delete and display student records with file-based persistent storage.    | 07 Aug 2026 | Completed — submitted on track |
| 2    | Bank Management Application             | C++ application simulating deposit, withdrawal and balance-check operations using OOP and file handling.       | 14 Aug 2026 | Completed — submitted on track |
| 3    | Library Management System               | System managing books, members and borrowing records using structures/classes, with search by title or author. | 21 Aug 2026 | Completed — submitted on track |
| 4    | Mini Game Project (Tic Tac Toe / Snake) | Console-based game demonstrating loops, arrays and conditional logic, with win/loss detection and replay.      | 28 Aug 2026 | Completed — submitted on track |

Each task's submission was reviewed periodically by industry mentors, and overall completion of the internship was determined by the on-time, successful submission of all four projects, as confirmed by the task-tracking dashboard shown in Section 4.10.

## 4.9 Expected Outcomes

- Practical command of C++ syntax, control flow, and program structure.
- Applied understanding of object-oriented programming: classes, objects, and encapsulation.
- Hands-on experience with file handling for persistent data storage.
- Ability to design clear, menu-driven console interfaces.
- Skill in using arrays and structures to organize and manage real-world records.
- Capability to build, test, and debug complete console applications end-to-end.
- Successful, on-track completion and submission of all four assigned projects.

## 4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter issued by Thiranex, confirming the internship role, duration, and terms, is reproduced in Chapter 3 of this report. The task-tracking dashboard maintained by Thiranex confirms the on-time submission and successful completion of all four assigned C++ programming projects, as summarized below.

| Task | Project                   | Status                 | Submission                       |
|------|---------------------------|------------------------|----------------------------------|
| 1    | Student Management System | Completed Successfully | Submitted on track — 07 Aug 2026 |



|   |                                         |                        |                                  |
|---|-----------------------------------------|------------------------|----------------------------------|
| 2 | Bank Management Application             | Completed Successfully | Submitted on track — 14 Aug 2026 |
| 3 | Library Management System               | Completed Successfully | Submitted on track — 21 Aug 2026 |
| 4 | Mini Game Project (Tic Tac Toe / Snake) | Completed Successfully | Submitted on track — 28 Aug 2026 |

All four projects were marked "Completed Successfully" in the task-tracking dashboard, confirming full and on-time completion of the internship deliverables.

## 5. BIBLIOGRAPHY/REFERENCES

- Thiranex, "Offer Letter for Internship in C++ Programming," Intern ID: THX-JUL3126-376, 31 July 2026.
- Thiranex, official website, <https://www.thiranex.in>
- cppreference.com — C++ Language Reference, <https://en.cppreference.com/>
- GeeksforGeeks — Object-Oriented Programming in C++, <https://www.geeksforgeeks.org/>
- cplusplus.com — File Handling in C++, <https://cplusplus.com/doc/tutorial/files/>